

RESEARCH ARTICLE

An unsupervised subdomain adversarial network for remaining useful life estimation under various conditions

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Abstract

Domain adaptation is a transfer learning method that is widely applied for remaining useful life (RUL) prediction under different operating conditions. However, current RUL prediction methods based on domain adaptation only consider minimizing degradation feature differences, while ignoring the correlation between features and labels, which may result in significant errors in the mapping relationship between the two learned by the model, and thus reduce the accuracy of prediction. To address this issue, this paper proposes an unsupervised subdomain adversarial network (USDAN) for RUL estimation under various conditions. On the basis of traditional domain adversarial adaptation networks, the network uses temporal convolutional networks to construct the feature extractor to learn the long-term dependencies between degradation data, and a multilinear conditioning scheme is designed to combine the features and subdomain labels as input features to the domain discriminator. Finally, with backpropagation and gradient reversal, the feature extractor is prompted to generate cross-domain invariant features that are correlated with the labels, improving the prediction accuracy and enhancing the transferability of the network. In the experimental section, the public turbofan engine degradation dataset is used to validate the effectiveness of the network, and the comparison with other approaches demonstrates its advantages.

KEYWORDS

domain adaptation, multilinear conditioning, remaining useful life, subdomain adversarial network, temporal convolutional network

1 | INTRODUCTION

Fault prognosis and health management (PHM) aims to predict and manage the possible future risks of the system, reduce the maintenance guarantee cost, improve the work efficiency of the system, and enable the mechanical equipment to operate more safely and reliably. The realization of PHM requires much technical support, such as state monitoring, fault monitoring and diagnosis, prediction, operation, and maintenance optimization.¹ RUL prediction is an important technical means of PHM, which can estimate the time from the current state to the failure according to the historical state data of the mechanical equipment operation and the failure mechanism.² The application of RUL can timely repair the upcoming faulty equipment, greatly reduce the loss caused by equipment failure, and enhance the safety and reliability of the system operation. However, the accuracy of RUL prediction will be affected by various factors, such as changes in

the equipment operating environment, noise during data collection, incomplete historical data, and so on.³ To strengthen the performance of RUL prediction, researchers have proposed many RUL prediction methods combined with various cutting-edge technologies, making RUL prediction a popular research direction in the field of PHM.

In recent years, with the rapid development and widespread application of artificial intelligence (AI), predicting RUL using AI technology has become increasingly popular, and such prediction methods can also be called data-driven methods. Compared with traditional model-based methods, data-driven methods reduce the dependence on prior knowledge and require no researchers to understand the operation and failure mechanism of mechanical equipment before prediction, which reduces the research difficulty. Moreover, the data-driven method makes it easy to extract the hidden features of the data, represent the degradation states of different operation stages more comprehensively, and establish an accurate mapping relationship with the labels. In the early stage of the study, machine learning was widely used in RUL prediction, such as artificial neural network (ANN),^{4,5} support vector machine regression (SVR),^{6,7} hidden Markov model (HMM),⁸ and so forth. However, since most of these methods are artificial to extract features, this requires researchers to learn relevant prior knowledge. With the rise of deep learning, its application in RUL prediction direction is now a hot topic because deep learning can completely get rid of the dependence on prior knowledge and has strong feature learning ability.⁹ A gating graph convolutional network for RUL point prediction and interval prediction that can fuse multisensor signals and extract temporal and spatial-dependent information was proposed by Wang et al.¹⁰ Jin et al. developed an RUL prediction method based on convolutional neural network (CNN) and location coding, which enhanced the enhancement of CNN's ability to learn sequence information.¹¹ Because the recurrent neural network (RNN) has great advantages for temporal data modeling, it can make up for the defect of CNN's inability to learn long-term time-dependent information. Therefore, in recent years, RNN-based methods and their combination methods with CNN have emerged in the field of RUL prediction.^{12–15} Wilberforce et al. studied a method combining CNN and RNN for predicting RUL in cells and developed a dropout method combined with callback techniques to reduce the risk of overfitting.¹⁶ Thakkar et al. proposed an RUL estimation technique based on deep RNN to calculate the RUL of turbofan engines based on complete historical data.¹⁷ Li et al. made the first attempt to use event vision data collected by event-based cameras for intelligent mechanical fault diagnosis designed an event data augmentation method to improve the robustness of the model, and further proposed a deep representation clustering method to improve the fault pattern identification of different machines.¹⁸ Zhou et al. developed a dual-thread gated recurrent unit (GRU) that uses a two-thread learning strategy to mine information in data and state differences between adjacent data, thus improving the performance of RUL prediction.¹⁹ However, deep learning methods fail to accurately predict test data different from the data distribution in the training set. Due to the changeable actual operation conditions of mechanical equipment, it is common that the existing historical data distribution is collected from different working conditions.

In order to solve the problem of reducing the prediction accuracy due to the difference in data distribution, the combination of transfer learning technology and RUL prediction is the main trend of current research. Transfer learning can use appropriate methods to transfer the knowledge learned in the source domain to the target domain to complete some tasks, such as pretraining and fine-tuning, sample weight transfer, domain adaptation, and so forth. Zhang et al. proposed a federated transfer learning approach for combining data privacy of different clients to complete mechanical fault diagnosis and used prior distribution to indirectly narrow the domain discrepancies.²⁰ Among them, the domain adaptation approach is the most widely used in the field of RUL prediction,^{21–23} which aims to learn the cross-domain invariance of features by narrowing the feature difference between the source and the target, so that the target domain data can be directly applied to the prediction model trained by the source dataset. Cheng et al. designed a transferable CNN using a multikernel maximum mean difference method to predict the RUL of bearings in multiple failure modes.²⁴ Miao et al. proposed a sparse domain adaptation network to improve the feature learning ability of CNN in the cross-domain RUL prediction tasks of bearings.²⁵ Hu et al. designed a weighted domain based on Wasserstein distance for the RUL estimation of bearings, addressing the negative effects of low-quality samples with noise and outliers on feature extraction.²⁶ A model combining CNN and adversarial-based domain adaptation method to predict the RUL of bearings running under different conditions and platforms was developed by Zhao et al.²⁷ Mao et al. proposed a self-supervised deep tensor domain adversarial regression adaptation method for online residual lifetime prediction of different machines to seek the optimal core tensor and domain invariant representation and achieve fast adaptive RUL predictions.²⁸ Zhuang et al. studied a metric adversarial domain adaptation method to predict the RUL of bearings under multiple working conditions, further learning the domain invariance of degradation features.²⁹ Ragab et al. developed a contrastive adversarial domain adaptation approach, incorporating contrastive losses based on the adversarial domain adaptation architecture and being able to consider target-specific information when learning domain invariant features.³⁰ However, there are still some problems with the above domain adaptation-based RUL prediction approaches affecting the prediction performance, which only focus

on minimizing the overall difference of degradation features, while ignoring the correlation between features and labels, which may result in significant errors in the learned mapping relationship between domain invariant features and labels in the target domain, and reduce the accuracy of prediction. To address this problem, this paper proposes an unsupervised subdomain adversarial network (USDAN) for RUL estimation under various conditions. First, a feature extraction network composed of temporal convolutional networks (TCNs) is used to extract the degeneration features of the degradation data, capturing the temporal dependence between the data. Then, the entire life cycle is divided into multiple segments to create a subdomain classification network, pretrained using the source data. The degeneration features of the source and target domain are combined with their classification labels to obtain new hidden features, which are input to the domain discriminator to train its ability to distinguish between the source and target domains. During the model optimization process, the feature extraction network is confused by the gradient reversal layer (GRL) so that the extracted degradation features of the source and the target are gradually similar, and then the cross-domain invariant features are obtained. Finally, the cross-domain invariant features of the target domain are input to the regression prediction network trained by the source domain to obtain the corresponding RUL labels. The main contributions of this paper are summarized below.

- (1) The proposed USDAN is based on the traditional domain adversarial adaptation network (DAAN) and designs a multilinear conditioning method to combine features and labels as the input features of the domain discriminator, improving the simplicity of the mapping relationship between domain invariant features and lifetime labels, and enhancing the transferability of the network.
- (2) The proposed framework uses TCNs to compose feature extraction modules to learn the cross-domain invariance of degradation features in the source and target domain, which can both extract spatial features to the data and capture long-term time-dependent information of the sequence data.
- (3) This paper uses the turbofan engine simulation degradation datasets to design experiments to verify the effectiveness of the proposed USDAN.

The rest of this article is arranged as follows. The second section describes the problems to be addressed in this paper and the underlying methods used. The third section introduces the specific implementation of the network architecture and method proposed in detail. The fourth part expounds on the experimental process and the result analysis. The last section summarizes the article and puts forward the next research direction.

2 | METHODOLOGY

2.1 | Problem description

RUL prediction uses the domain adaptation method to solve two main problems: (1) The distribution of the training set and the test set data is inconsistent, and the RUL of the prediction test data of the network model optimized by the training set is easy to produce serious errors. (2) The historical data of the test set have no label and cannot be directly used to train the prediction model. Therefore, the domain adaptation-based RUL prediction method is an unsupervised learning method. Given the sensor data collected from a variety of operating conditions, including the labeled source dataset $SD = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ and the unlabeled target dataset $TD = \{x_i^t\}_{i=1}^{n_t}$, following the different data distributions P and Q , respectively. It should be noted that SD and TD are collected from mechanical components of the same type and quality, with the same maximum useful life and label space.

The current prediction algorithms based on domain adaptation are generally divided into three steps. In the first step, the feature generation network N_f was constructed to extract the feature representation of the source domain and the target domain, and the features of the i th time step are expressed as $f_i^s = N_f(x_i^s)$ and $f_i^t = N_f(x_i^t)$. In the second step, the feature distribution differences are narrowed so that the feature extraction network extracts cross-domain invariant feature representations, namely $f_i^s \approx f_i^t$. In the third step, predict the RUL of the target domain, denoted $y_i^t = N_p(f_i^t)$. However, as shown in Figure 1, these methods have a problem that cannot be ignored. The extracted deep feature representation is insufficient and may lead to domain mismatch, resulting in incorrect alignment of degradation features with different RUL labels, $f_m^t \approx f_n^t, m \neq n$. On the whole, RUL estimation belongs to the category of the regression problem, and the necessary link to solve this problem is to establish a one-to-one functional mapping relationship between features and labels. However, the occurrence of the above problems will increase errors in the mapping between features and labels and reduce the accuracy of RUL estimation. This paper focuses on proposing a solution to this problem.

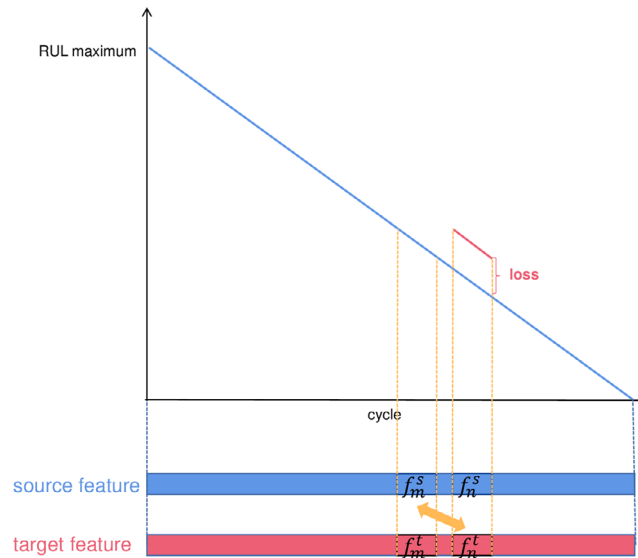


FIGURE 1 Schematic diagram of the domain mismatch.

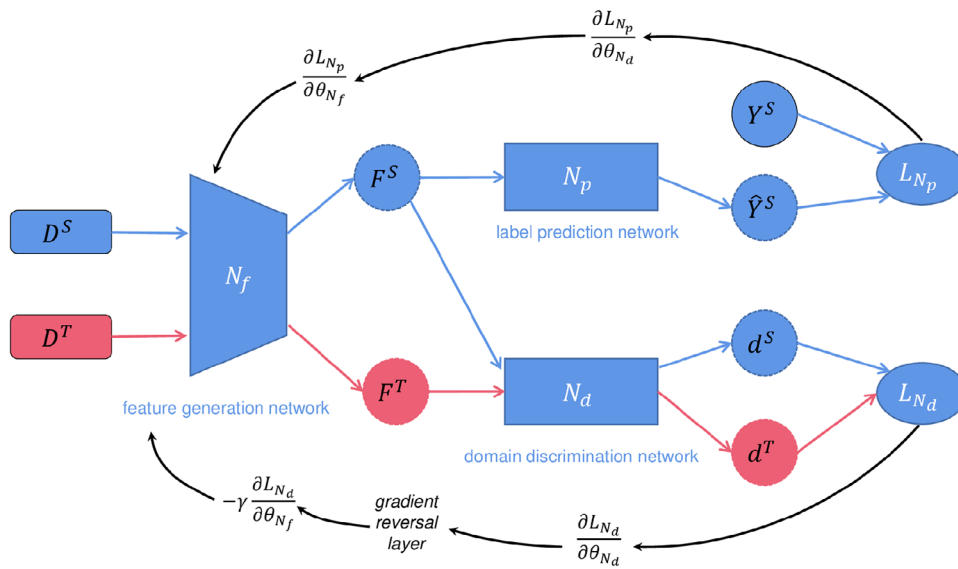


FIGURE 2 Network structural diagram of DAAN.

2.2 | Domain adversarial adaptation network

As one of the most widely used methods of domain adaptation, the core idea of the DAAN³¹ is to narrow the discrepancy between edge distributions $P(x)$ and $Q(x)$ from the level of feature representation. As shown in Figure 2, DAAN consists of three parts, including the feature generation network N_f , domain discrimination network N_d , and label prediction network N_p . There is an important component between N_f and N_d , the GRL, which is also the key to extracting the feature invariance. The GRL has no effect on the input of the features during forward propagation. On backpropagation, the gradient propagating forward from N_d passes the GRL into the opposite number after multiplying with a certain coefficient, specifically expressed as follows:

$$\frac{\partial L_{N_d}}{\partial \theta_{N_d}} \xrightarrow{\text{GRL}} -\gamma \frac{\partial L_{N_d}}{\partial \theta_{N_f}} \quad (1)$$

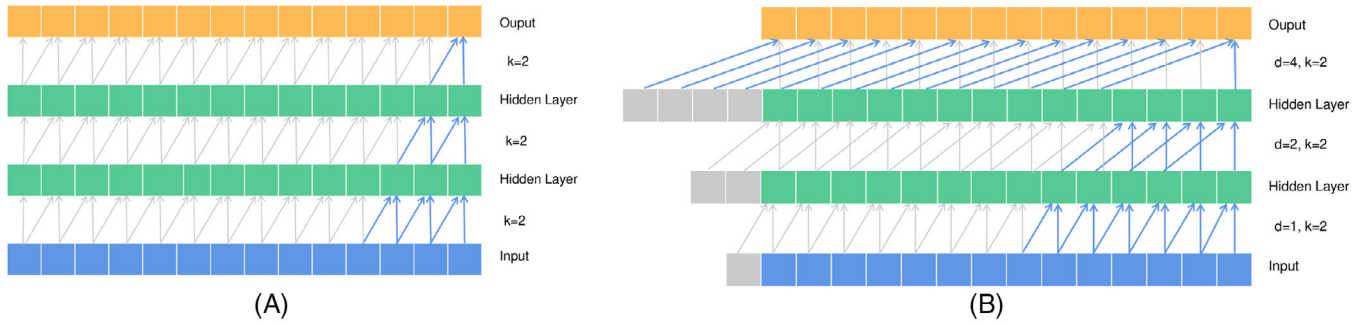


FIGURE 3 Schematic of the convolution operation. (A) Causal convolution, (B) dilated causal convolution.

where, L_{N_d} represents the discriminator loss function. N_d is optimized in the direction of decrease, while after GRL, the gradient of N_f is optimized in the direction of L_{N_d} increase. In this way, an adversarial relationship between the two is formed. The final result of training is that N_f can capture the feature invariance of the domain, and N_d cannot distinguish which domain the input feature comes from. Thus, the objective function of DAAN is a maximum–minimum function,

$$L_{N_d}(\theta_{N_d}) = \max_{\theta_{N_d}} E_{x^s \in SD} \log(N_f(x^s)) + E_{x^t \in TD} \log(N_f(x^t)) \quad (2)$$

$$L_{DAAN}(\theta_{N_f}, \theta_{N_d}, \theta_{N_p}) = \min_{\theta_{N_f}, \theta_{N_d}, \theta_{N_p}} E_{(x^s, y^s) \in SD} L_{N_p}(N_p(N_f(x^t), y^s)) + \gamma L_{N_d}(\theta_{N_d}) \quad (3)$$

2.3 | Temporal convolutional network

TCN³² is a kind of CNNs used to solve one-dimensional time series problems. RNNs are generally used to model timing problems because the recurrent auto-regressive structure of RNNs can well capture the long- and short-term time dependence. However, traditional CNNs cannot grasp the information in the long-term time due to the size of their convolutional kernel, and the emergence of TCN solves this problem. At the same time, TCN also avoids the common problems of RNN, such as gradient explosion or disappearance, and poor long-term memory. In order not to disrupt the temporal order of the sequence data, the input and output length of the TCN adjacent convolution layers must remain consistent. Causal convolution is a key technique for TCN implementation to capture time-dependent information such as RNN, as shown in Figure 3A. The core idea of causal convolution is that the element of any position i of the sequence species is influenced only by its previous elements, but not by the latter elements. It should be noted that in order to ensure the same input and output length, the left side of the dimension of the input data requires a zero complement operation so that there are enough elements at the end of the sequence for convolution operation to generate output features of the same length.

However, the causal convolution is limited by the size of the convolution kernel. When the length of the input sequence is long, TCN needs to use many layers of convolutional networks to capture longer dependence information, and the learnable weight parameters increase accordingly. To solve this problem, the dilation convolution technique is proposed in TCN, and the implementation principle is shown in Figure 3B. Unlike traditional convolution, dilation convolution allows the input to have interval sampling, and the length of the interval is controlled by the sampling rate, namely the dilation rate. In general, the sampling rate is larger as the number of convolution layers increases. Thus, dilated convolution can allow TCN to use fewer convolution layers to obtain larger receptive fields and capture longer time-dependent information.

In order to optimize the training of deep networks in TCN, residual connections are applied to TCN. Residual is a commonly used deep learning optimization technique that adds the network output to the original input as the final output. The residual connection only contains a one-dimensional convolutional layer, which changes the number of input data channels to ensure consistency with the number of network output channels for easy addition. As indicated in Figure 4, the construction of time convolutional blocks is to replace the dilation causal convolutional layers in the original TCN with a residual block, which generally contains two one-dimensional convolutional layers with the same dilation rate and adds residual connections. In addition, to avoid the risk of gradient explosion and overfitting, normalization and dropout layers need to be added to the residual block after the expansion causal convolution layer.

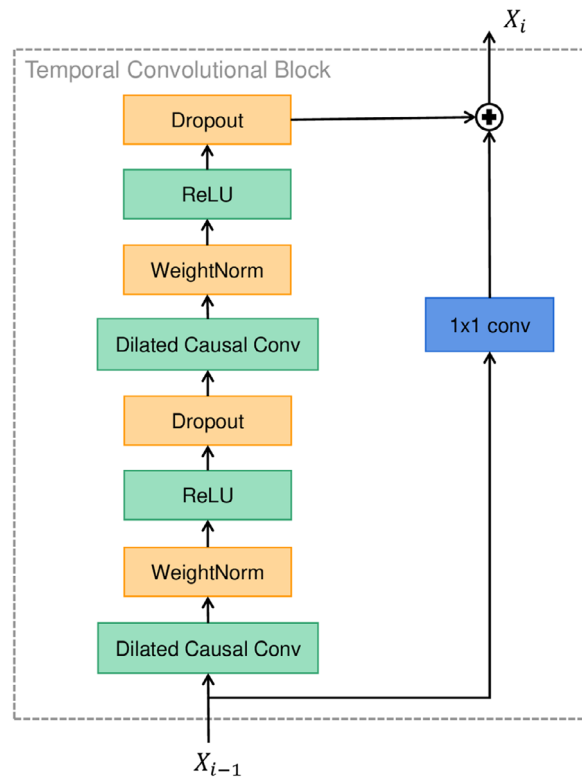


FIGURE 4 Structural diagram of the temporal convolutional block.

3 | UNSUPERVISED SUBDOMAIN ADVERSARIAL NETWORK WITH TEMPORAL CONVOLUTIONAL NETWORK

To address the problem of domain mismatch caused by inadequate deep feature representation extraction in domain adaptation, we propose an USDAN. The network framework is based on the basic structure of the DAAN and consists of four parts, feature generation network N_f , subdomain clustering network N_{sub} , domain discrimination network N_d , and label prediction network N_p . N_f is built by TCN as infrastructure and is used to extract domain-invariant features of the source and target domain. N_d is a binary classification network used to distinguish the sources of degradation features of the input. The input of N_p is the domain-invariant feature sent from the feature generation network, and the output is the RUL label. In this section, we will focus on the implementation of the domain identification network, which is different from the traditional DAAN and is also the innovation point of this paper. Traditional DAAN directly feeds the degradation features generated by N_f into N_d to identify the source and then uses the GRL cheat N_f during backpropagation, making the extracted features of the source and target domain more and more similar. However, these methods all ignore the correlation between features and labels. No labels restrict the generation of invariant features in the domain, which may lead to insufficient feature representation and no uniqueness. This can cause a serious error, namely, a domain mismatch. Domain mismatch means that domain-invariant features at different time steps will be mapped to the same RUL prediction labels, resulting in greatly reduced transfer results and prediction performance.

Inspired by the conditional adversarial domain adaptation,^{33,34} this paper develops an unsupervised subdomain adversarial method, the core idea is to divide the discrete label space into multiple subdomains, and then combine the subdomain category labels and features into a new feature as input to the domain discriminating network. This method links domain-invariant features to subdomain labels, so that the features fed into have characteristics distinct from those of other time periods, and more fully represent the state of the current degeneration period. Next, we will introduce the specific implementation of the method.

3.1 | Subdomain division

First, the RUL label space is divided into n_{sub} subdomain spaces, and each subdomain space corresponds to a classification. The division is based on the whole life cycle from the initial work to the failure into n subsegments, different segments can represent different degrees of degradation. The subdomain label space is calculated as follows:

$$Y_{sub} = \text{unique} \left(\left\{ y_{sub}^i = \text{round} \left(\frac{RUL_{\max} - t}{RUL_{\max}} \times n_{sub} \right) \mid 0 \leq t \leq RUL_{\max} \right\} \right) \quad (4)$$

where $RUL_{\max}$ denotes the maximum remaining useful life (RUL), $\text{unique}(\cdot)$ is a function that removes the duplicate values in the set and returns the processed set, and $\text{round}(\cdot)$ is a rounding function. The labeled source domain data calculate the subdomain category for all time steps using (4). A subdomain classification network was constructed using the source domain data as the training set and the features generated by N_f as input. The network calculates the subdomain classification error using the cross-entropy loss function with the following formula:

$$L_{sub} = \frac{1}{n_s} \sum_{i=1}^{n_s} (-y_{sub}^i \log \hat{y}_{sub}^i - (1 - y_{sub}^i) \log (1 - \hat{y}_{sub}^i)) \quad (5)$$

where n_s represents the number of training samples of the source domain, y_{sub}^i represents the true subdomain label of the i th sample, and $\hat{y}_{sub}^i$ represents the subdomain prediction label of the i th sample. Backpropagation techniques are used to optimize and update the parameters and weights of the subdomain classification networks. The target degradation features generated by N_f are input into the pretrained subdomain clustering network to obtain the subdomain category pseudo-label $\hat{Y}_{sub}^T$.

3.2 | Multilinear conditioning

Multilinear conditioning aims to combine degeneration features with their subdomain labels using the method of multilinear mapping. The advantage of multilinear mapping is to simulate the multiplicative interactions between different variables, which can express the intrinsic connection between labels and multisensor data more than the joint method of element-level addition and head-to-end linkage. The new hidden features H_i^s and H_i^t of the source and target domain are calculated as follows:

$$H_i^s = f_i^s \otimes y_{sub(c)}^s \quad (6)$$

$$H_i^t = f_i^t \otimes \hat{y}_{sub(c)}^t \quad (7)$$

where f_i^s and f_i^t represent the degeneration features of the i th time step of the source and target domain extracted by N_f , $y_{sub(c)}^s$ denotes the category label of the current time step of the source domain belonging to the subdomain c , and $\hat{y}_{sub(c)}^t$ denotes the pseudo category label of the current time step of the target domain belonging to the subdomain c .

3.3 | Joint objective function

To achieve the discrimination of new hidden features with multilinearly conditioning for the source and target, the binary classification cross-entropy loss function was used to compute the classification error of N_d . The formula was calculated as indicated in Equation (8). Furthermore, to establish accurate mapping function relationships between degradation features and RUL labels, the prediction error of N_p is computed using the root mean square loss function. The formula was calculated as indicated in Equation (9).

$$L_{N_d} = -y_d^i \log N_d(H_i) - (1 - y_d^i) \log (1 - N_d(H_i)) \quad (8)$$

$$L_{N_p} = \sqrt{\frac{1}{n_s} \sum_{i=1}^{n_s} |y_i - N_p(f_i)|^2} \quad (9)$$

where y_d^i denotes the actual category label of the i th new hidden feature belonging to the source or the target, and y_i denotes the real RUL label of the source domain. Therefore, the total objective function of the USDAN network consists of three parts, subdomain classification objective function, domain discrimination objective function, and label prediction objective function, which are expressed as follows:

$$L_{USDAN}(\theta_{N_f}, \theta_{N_p}, \theta_{N_d}, \theta_{N_{sub}}) = L_{N_p}(\theta_{N_p}) + \mu L_{N_{sub}}(\theta_{N_{sub}}) + \gamma L_{N_d}(\theta_{N_d}) \quad (10)$$

where, θ_{N_f} , θ_{N_p} , θ_{N_d} , and $\theta_{N_{sub}}$ denote the parameters of the four subnetworks of USDAN, respectively, and $\mu = 2/(1 + e^{-\frac{\epsilon}{\max_iter}}) - 1$, $r = 2/(1 + e^{-\frac{\epsilon}{\max_iter}}) - 1 \in [0, 1]$, are the time-varying coefficient, which gradually increase with the training iteration. ϵ is a constant, $iter$ and $\max_iter$ are the current iterations number and the total number of iterations, and $batch$ and $\max_batch$ are the current batch id and the total batch number.

In order to update and optimize the weights and parameters of the USDAN model, the stochastic gradient descent (SGD) algorithm is used to calculate the weights during the backpropagation process and determine the direction of gradient optimization. The model parameters are updated as shown below, where δ denotes the learning rate.

$$\left\{ \begin{array}{l} \theta_{N_f} \leftarrow \theta_{N_f} - \delta \frac{\partial L_{N_p}(\theta_{N_p}) + L_{N_{sub}}(\theta_{N_{sub}}) - L_{N_d}(\theta_{N_d})}{\partial \theta_{N_f}} \\ \theta_{N_p} \leftarrow \theta_{N_p} - \delta \frac{\partial L_{N_p}}{\partial \theta_{N_p}} \\ \theta_{N_d} \leftarrow \theta_{N_d} - \delta \frac{\partial L_{N_d}}{\partial \theta_{N_d}} \\ \theta_{N_{sub}} \leftarrow \theta_{N_{sub}} - \delta \frac{\partial L_{N_{sub}}}{\partial \theta_{N_{sub}}} \end{array} \right. \quad (11)$$

3.4 | RUL prediction based on USDAN

This paper presents a novel transfer learning network USDAN for RUL prediction to solve the problem of accuracy degradation due to domain mismatch. The RUL prediction process for this network is as follows and the pseudo-code of the proposed method is expressed in Algorithm 1.

- (1) The training data and testing data are collected using the sensor monitoring target device, where the labeled data are used as the source domain and the unlabeled data are used as the target domain.
- (2) All degradation data are preprocessed using normalization operations and sliding time window techniques, unifying the magnitude of the data items and expanding the characteristic data for a single time step. Later, the preprocessed source and target domain data are input into the feature generation network to obtain degradation features. Next, the degeneration features propagate forward to the subdomain clustering network to generate their subdomain labels and carry out multilinear conditioning with the feature tensors to obtain new hidden features. N_d receives new hidden features and distinguishes their sources. Meanwhile, degradation features serve as input to N_p to generate RUL prediction labels.
- (3) The error loss of the generated and true labels for each network is calculated, and the SGD backpropagation is used to update the parameters and optimize the model. According to the characteristics of DAAN, N_f is deceived due to the gradient reverse layer, causing the extracted features to develop in the direction of increasing the discrimination error of N_d .
- (4) The ultimate result is that N_f can capture the domain invariance of the source features and the target features, making N_d unable to distinguish the sources of input features. This means that the optimization of the network is complete. The extracted domain-invariant features are fed into N_p to obtain the final RUL prediction labels.

ALGORITHM 1 The training and prediction of USDAN model.

Input: $D_S = \left\{ \left(x_j^s, y_j^s \right) \right\}_{j=1}^{n_s}$, $D_T = \left\{ x_j^t \right\}_{j=1}^{n_t}$, batch size = n_{batch}

- 1: Preprocess the original sensor data of the source domain and target domain using maximum and minimum normalization and sliding time window techniques;
- 2: Divide the whole life cycle into n_{sub} subdomains and calculate the true subdomain label $C_{sub}^s = \{c_i^s\}_{i=1}^{n_s}$ of the source domain;
- 3: Initialize the parameters of USDAN model;
- 4: // USDAN model training
- 5: **for** $epoch = 1$ to Max_epoch **do**
- 6: **for** $i = 1$ to Max_batch **do**
- 7: // Extract degradation feature
- 8: Use TCN to generate source feature f_i^s ;
- 9: Use TCN to generate target feature f_i^t ;
- 10: //Regression and classification
- 11: Calculate source prediction output $\hat{y}_i^s$ and subdomain classification output c_i^s ;
- 12: Calculate target prediction output $\hat{y}_i^t$ and subdomain classification output $\hat{c}_i^t$;
- 13: //Use multi-linear mapping to combine feature and subdomain label
- 14: Calculate source new hidden feature $h_i^s = f_i^s \otimes c_i^s$;
- 15: Calculate target new hidden feature $h_i^t = f_i^t \otimes \hat{c}_i^t$;
- 16: Fed h_i^s and h_i^t into domain discrimination network;
- 17: // Calculate loss and gradient descent
- 18: Calculate the classification error $L_{N_{sub}}$;
- 19: Calculate the discrimination error L_{N_d} ;
- 20: Calculate the prediction error L_{N_p} ;
- 21: Calculate the total loss $L_{USDAN} = L_{N_p} + \mu L_{N_{sub}} + \gamma L_{N_d}$;
- 22: Backpropagation and update the parameters;
- 23: **end for**
- 24: **end for**
- 25: Use the trained USDAN model to predict the RUL value of target testing dataset.

Output: Weights and biases $\{W, b\}$ of USDAN model, RUL prediction value of target data.

TABLE 1 Description of the C-MAPSS dataset.

Sub dataset	Training units	Testing units	Operating conditions	Fault modes
FD001	100	100	1	1
FD002	260	259	6	1
FD003	100	100	1	2
FD004	249	248	6	2

4 | EXPERIMENT

4.1 | Dataset description and preprocessing

This study uses a commercial Modular Aviation Propulsion System Simulation (C-MAPSS) dataset from NASA,³⁵ which is a widely used benchmark dataset. The experimental engine is shown in Figure 5. C-MAPSS includes different amounts of sensor data for operating conditions and fault modes. As shown in Table 1, this data set has four subdatasets, each of which is divided into a training subset and a testing subset, where the training set records the sampling value of each time series of multiple state parameters of the aero-engine from normal to failure, and the testing set contains the state parameter values before a time point before the failure and the corresponding remaining life. Each row in the data is a snapshot of the data over a single running time period, containing 26 columns, from left to right representing the engine

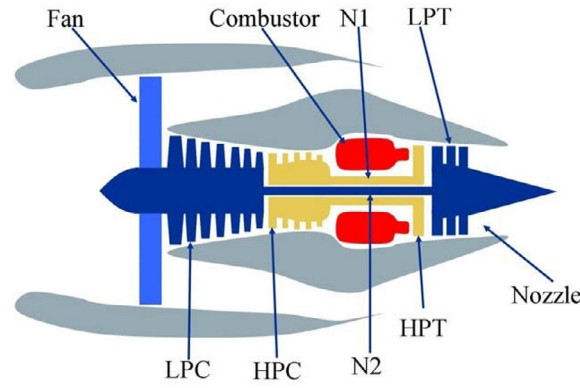


FIGURE 5 Schematic diagram of the engine.

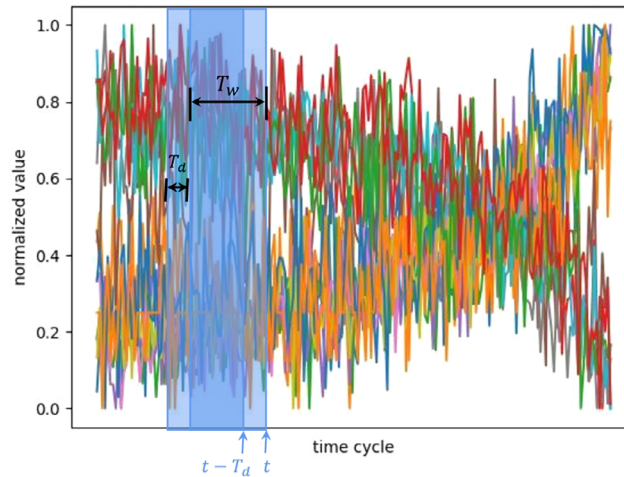


FIGURE 6 Implementation diagram of sliding time window technology.

ID, the current number of operating cycles, three operating settings that have a significant impact on engine performance, and 21 sensor values. The engine worked properly at the start of each time series but failed at some unknown point in time. In the training set, the degree of failure grows over time until the system fails. In the testing set, the data are provided until some time before the system fails. The goal of RUL prediction is to estimate the remaining lifetime of the testing system before it fails.

The state parameters monitored by the 21 sensors do not all effectively reflect the degradation state of the current engine. A closer observation of the dataset shows that there are some sensors whose monitoring values change very little from the engine running to the failure. These invalid sensor values should be removed from the degradation data to reduce the computational cost of the prediction model. Thus, there are 14 sensors available as input data. However, the numerical magnitude of these sensors varies somewhat, which can affect the accuracy and convergence rate of the model. Therefore, the maximum minimum normalization method was used to unify all sensor values in the degraded data within [0,1], with the following formula.

$$\hat{x}_i^{sn} = 2 \frac{\max(x^{sn}) - x_i^{sn}}{\max(x^{sn}) - \min(x^{sn})} \quad (12)$$

where x_i^{sn} denotes the monitored value of the sensor numbered in the i th time cycle, $\max(x^{sn})$ and $\min(x^{sn})$ indicate the maximum and minimum values of the sensor throughout the operation cycle, and $\hat{x}_i^{sn}$ represents the normalized data. Since there are only 14 state parameters in each time cycle, it is impossible to fully represent the running state of the engine in the current cycle. According to previous research,³⁶ the maximum service life of all engine units in the C-MAPSS dataset can be set to 125. Then, this study used sliding time window technology to amplify characteristic data for all time periods on the basis of existing historical data. A specific implementation of this technique is shown in Figure 6. Where, T_w and

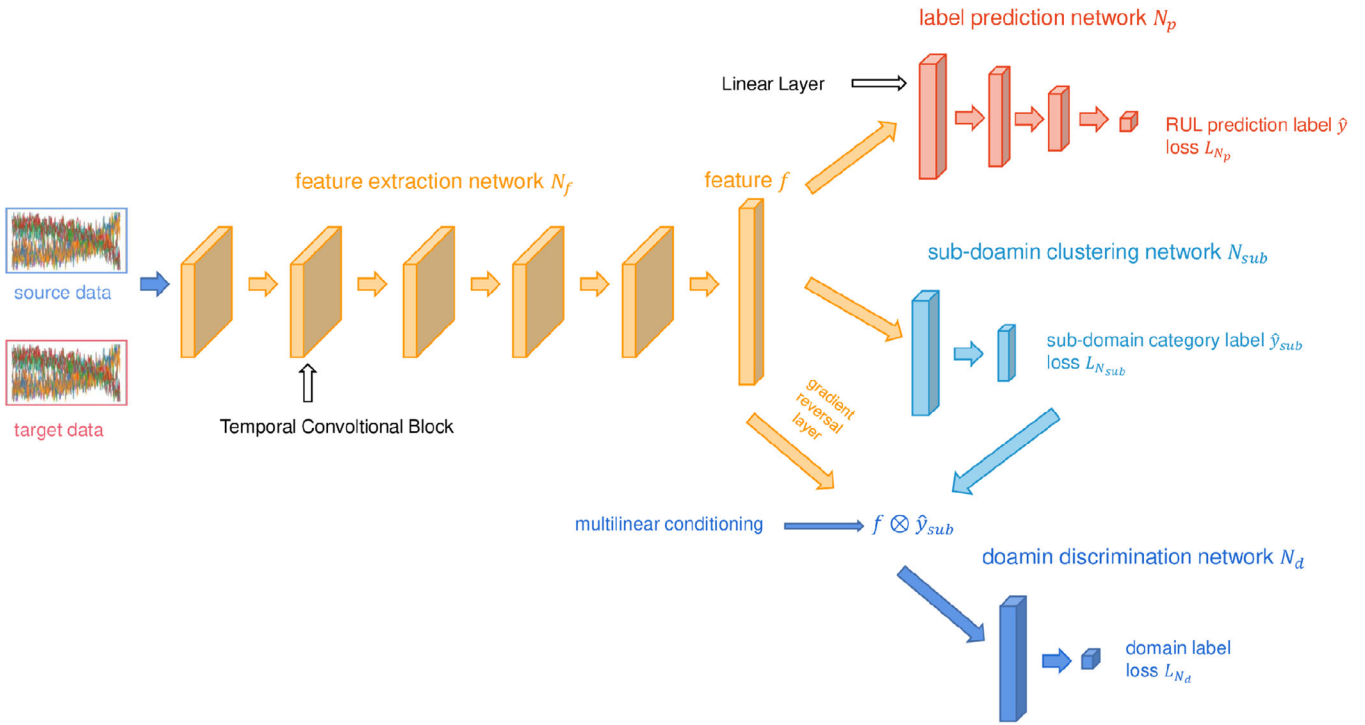


FIGURE 7 The network structure diagram of the USDAN.

T_d indicate the sliding window size and the sliding time step, respectively. The section covered by the blue box indicates the feature data at the current time step t .

4.2 | Model configuration and training

Figure 7 illustrates the frame structure of the USDAN. N_f consists of five temporal convolution blocks with a convolution kernel size of 3 and a consistent number of output channels for each block, which is 40. The dilation coefficient of the dilation casual convolution is a power of 2, at 1, 2, 4, 8, 16, respectively. It should be noted that TCN uses a dilation casual convolution to extract long-term time-dependent information of the data and to maintain consistent input–output lengths. In the process of convolution operation, the input data need to be supplemented with zeros on the left side to meet the generation of end data in the hidden features of the next layer. The supplementary number of zero is calculated as follows:

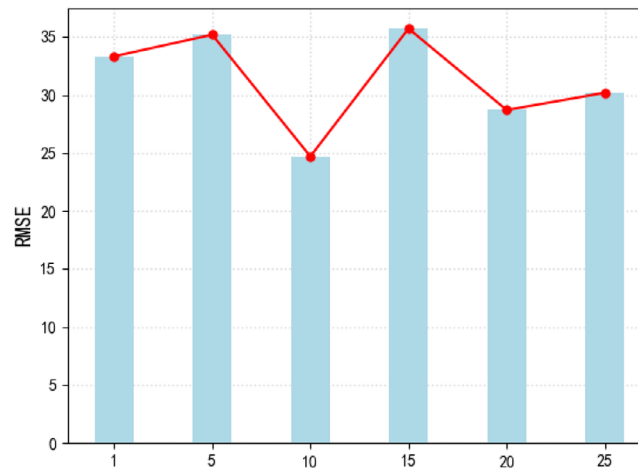
$$n_{zero}^i = (n_{ks} - 1) \times d_i \quad (13)$$

where n_{zero}^i and d_i represent the number of the filled zeros and the dilation rate of the i th hidden layer, and n_{ks} represents the size of the convolution kernel.

Moreover, because the convolution operation simultaneously fills the zero on both sides of the data, the cut layer is added to the temporal convolution block and placed behind the convolution layer to cut off the right-filled zero. The constituent components of the subdomain clustering network are in turn dropout layer, the linear layer with output dimension 100, the activation function layer, and the linear layer with output dimension 10. N_p contains one dropout layer and three linear layers, with an activation function layer placed after the first two linear layers, and the output dimension is half that of the previous linear layer. The output dimension of the last linear layer is 1. The structure of N_d is similar to the subdomain clustering network, with the only difference being that its final output dimension is 1. Since the input of N_d is a new hidden feature with multilinear conditioning, the size of its input dimensions is the product of the linear expansion dimension of degradation features generated by N_f and the number of subdomains. The activation functions used in USDAN are rectified linear units (ReLU). After multiple parameter tuning, we determined the values of hyperparameters in Table 2 to achieve the optimal performance of the model. The size of the convolution kernel of the temporal convolution block in N_f is 3, and the full life cycle is divided into ten sub-domains. In the

TABLE 2 Model hyperparameter settings for USDAN.

Hyperparameters	Values
Number of subdomain n_{sub}	10
Temporal convolutional kernel size n_{ks}	3
Batch size	32
Sliding window size T_w	30
Sliding time step T_d	1
Learning rate δ	0.0002
Max epoch	220

**FIGURE 8** Experimental results of different numbers of subdomains.

data preprocessing phase, the size of the sliding window is 30 and the sliding time step is set to 1. During the iterative training process, the batch size of the input data is 32, and the learning rate is $2e-4$. The maximum number of iterations is 220. It should be noted that the study focuses on exploring the impact of subdomain partition density on model prediction performance. We conducted experiments using different numbers of subdomains, and the experimental results are shown in Figure 8. It is obvious that when the number of subdomains is 10, the accuracy of the model is the highest, which also means that the density of subdomain partitioning is not positively correlated with model performance.

To validate the effectiveness of our proposed method, 12 sets of cross-domain transfer tasks were designed in this study, as shown in Table 3. The C-MAPSS dataset has four subdatasets, and each subdataset establishes a transfer task with the remaining subdatasets. To comprehensively evaluate the model performance as well as the experimental results of these transfer tasks, this study uses two performance evaluation metrics, root mean square error (RMSE) and mean absolute error (MAE).

$$RMSE = \sqrt{\frac{1}{n_t} \sum_{i=1}^{n_t} |\hat{y}_i - y_i|^2} \quad (14)$$

$$MAE = \frac{1}{n_t} \sum_{i=1}^{n_t} |\hat{y}_i - y_i| \quad (15)$$

where, $\hat{y}_i$ and y_i denote the prediction and true values, and n_t denotes the number of test samples in the target domain.

TABLE 3 Performance comparison of different transfer learning methods on various transfer tasks of C-MAPSS dataset.

Transfer task	Baseline		DAAN		Proposed	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
FD2-1	39.8	36.1	33.2	26.7	24.7	19.7
FD3-1	18.9	14.6	17.8	14.0	16.2	13.1
FD4-1	32.2	24.5	30.4	26.4	20.4	17.4
FD1-2	48.4	40.8	43.4	37.6	39.9	35.0
FD3-2	52.8	43.5	51.4	41.9	48.7	41.3
FD4-2	17.5	12.9	17.4	13.0	16.4	12.7
FD1-3	48.6	42.7	45.5	38.7	19.1	14.8
FD2-3	44.5	37.5	39.6	32.8	35.5	31.0
FD4-3	56.0	48.4	51.7	42.7	37.3	28.6
FD1-4	51.6	43.6	45.3	39.0	43.9	38.9
FD2-4	43.1	32.5	37.3	28.9	32.1	26.5
FD3-4	54.1	44.4	53.5	45.0	53.1	44.3
Mean	42.3	35.1	38.9	32.2	32.3	26.9

The bold values represent the best results.

4.3 | Method comparison and performance analysis

To highlight the strengths of our proposed method, this study compares and analyzes the performance with other methods based on 12 sets of experimental tasks. Since the C-MAPSS dataset contains degradation data collected under four different working conditions, we use the data of each condition as the target domain, and the data of other conditions as the source domain, which can design three sets of migration tasks. The comparison methods used in this paper are divided into three parts: the baseline method of not using transfer learning, the traditional DAAN method, and the other state-of-the-art transfer learning methods. These methods are described in detail below.

(1) Baseline

The baseline method uses the prediction model trained on the source domain to directly apply to the target domain and is not applicable to the transfer learning approach. In this paper, N_f and N_p in USDAN are used to construct the training model of the baseline method, using the source domain data for pretraining to predict the RUL of the target domain.

(2) DAAN

DAAN is a traditional adversarial-based domain adaptation method, which consists of three structures: feature extractor, label predictor, and domain discriminator. Unlike the USDAN method proposed in this paper, where the multilinear conditioning method is not used in DAAN, domain discriminators directly distinguish features derived from feature extractors. The purpose of this study as a comparison method is to verify the effect of multilinear conditioning on the performance of the prediction model.

(3) Other transfer learning methods

Recently, many transfer learning methods have been applied to RUL prediction. For example, Zhang et al.'s CNN-MMD and TL-DRR,³⁷ use a transfer learning prediction method for deep feature representation. Costa et al.'s LSTM-DAAN and CNN-DAAN³⁸ combine LSTM and CNN with domain adversarial neural networks to solve domain shifts in RUL prediction. In contrast to the USDAN proposed in this paper, domain adaptation networks (DAN) usually measure the difference between the source domain and the target domain using methods such as maximum mean difference or KL divergence, and obtain the domain invariant features of the source and target by minimizing the difference during training. The domain subdomain adaptation network (DSAN)³⁹ mainly divides the degradation data of the whole life cycle into multiple segments, with each segment representing a subdomain, and calculates and minimizes the difference between the source and target domain in each same domain to achieve the subdomain alignment.

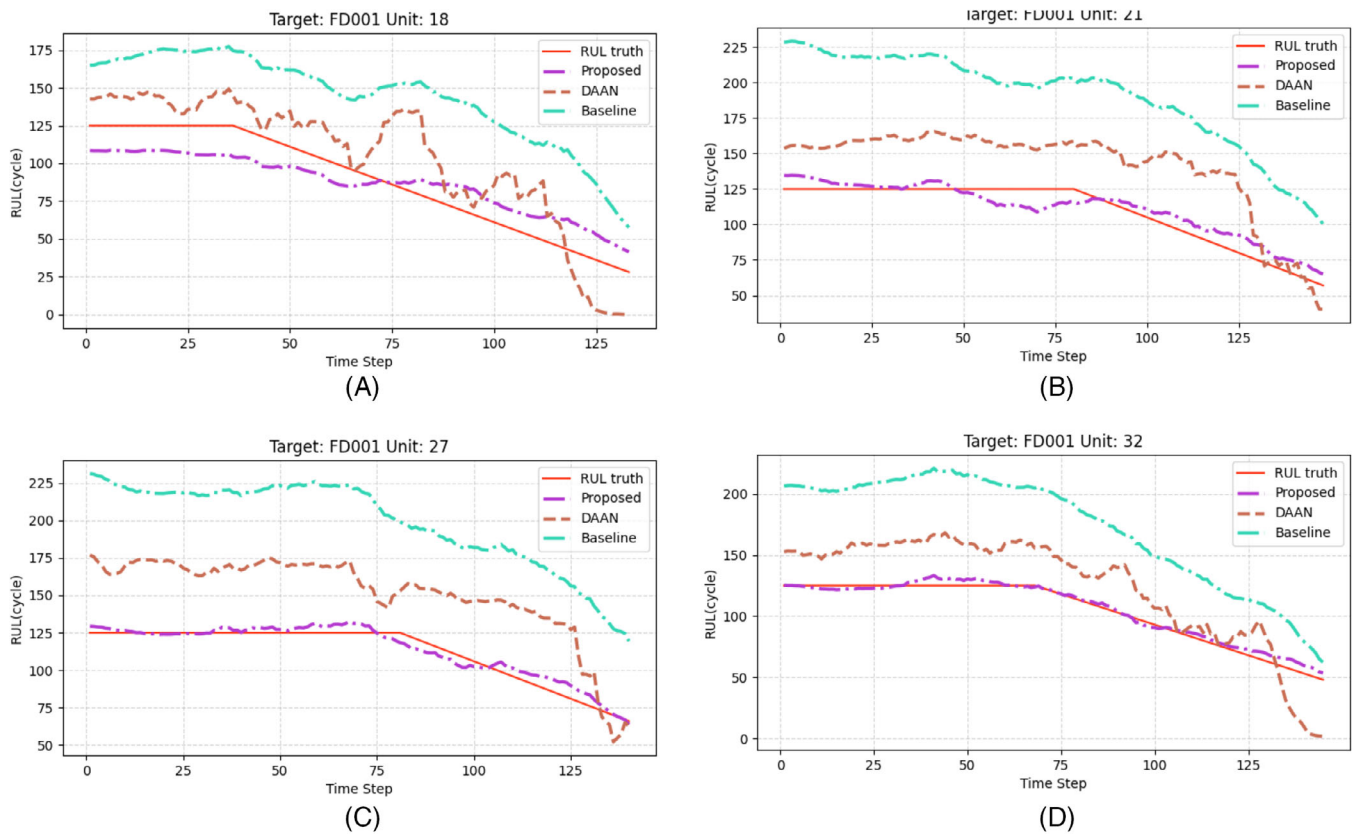


FIGURE 9 Prediction plots of different engine units with different methods on FD2-1.

This experiment performed the transfer learning task based on the C-MAPSS dataset using the USDAN, Baseline, and DAAN methods. Table 3 records the results of experiments containing two performance indicators. Intuitively, the proposed method has outperformed the Baseline and DAAN methods on these 12 sets of transfer tasks. From a statistical perspective, the mean RMSE of Baseline, DANN, and USDAN methods are 42.3, 38.9, and 32.3, respectively. Comparing these two RUL prediction methods with the same feature generation network, the RMSE reduction rates of USDAN are 31.0 and 20.4%. The mean values of MAE for these three methods are 35.1, 32.2, and 26.9, respectively. The MAE of USDAN decreased by 30.5 and 19.7% compared to the other two methods. Moreover, we regarded the cross-domain prediction task with the same target domain as a group, and it can be found that for the target domain with different operation settings and failure modes, our proposed methods all show superior accuracy than Baseline and DAAN. The comparative results of the USDAN and Baseline methods illustrate that applying the transfer learning approach based on domain adversarial to cross-domain RUL prediction tasks is effective and can enhance the accuracy of the prognosis. The comparative results of the USDAN and DAAN methods show that splitting the entire life cycle into multiple subdomains and generating new discrimination features for subdomain adversarial through multilinear mapping can help to improve the prognosis performance, proving the effectiveness of our proposed method.

Moreover, in order to more comprehensively evaluate the prognosis performance of USDAN, this experiment has drawn two line plots, one is the comparison diagram of the estimation and true values of Baseline, DAAN, and USDAN methods on different engine units, such as Figures 9 and 10, and the other is the RUL prognosis and true values of DAAN on all engine units in the same subdataset, such as Figure 11. Figures 9 and 10 depict the prediction results of these three methods of different engine units on the transfer task FD2-1 and FD1-3. From these figures, we can observe that the prediction curve of the USDAN method better fits the real curve with the best performance, DAAN performing the second, and Baseline performing the worst. This is also consistent with the results of our statistical analysis. Furthermore, by comparing the prediction curves of DAAN with our proposed method, it can be found that the estimation curve of our proposed method is smoother and has fewer fluctuations near the true curve. This also proves that the USDAN restricts subdomain invariant features with subdomain labels, reduces the possibility of domain mismatch to some extent, and makes the mapping relationship between degradation features and RUL labels more robust. In addition, Figure 11 depicts the test results of

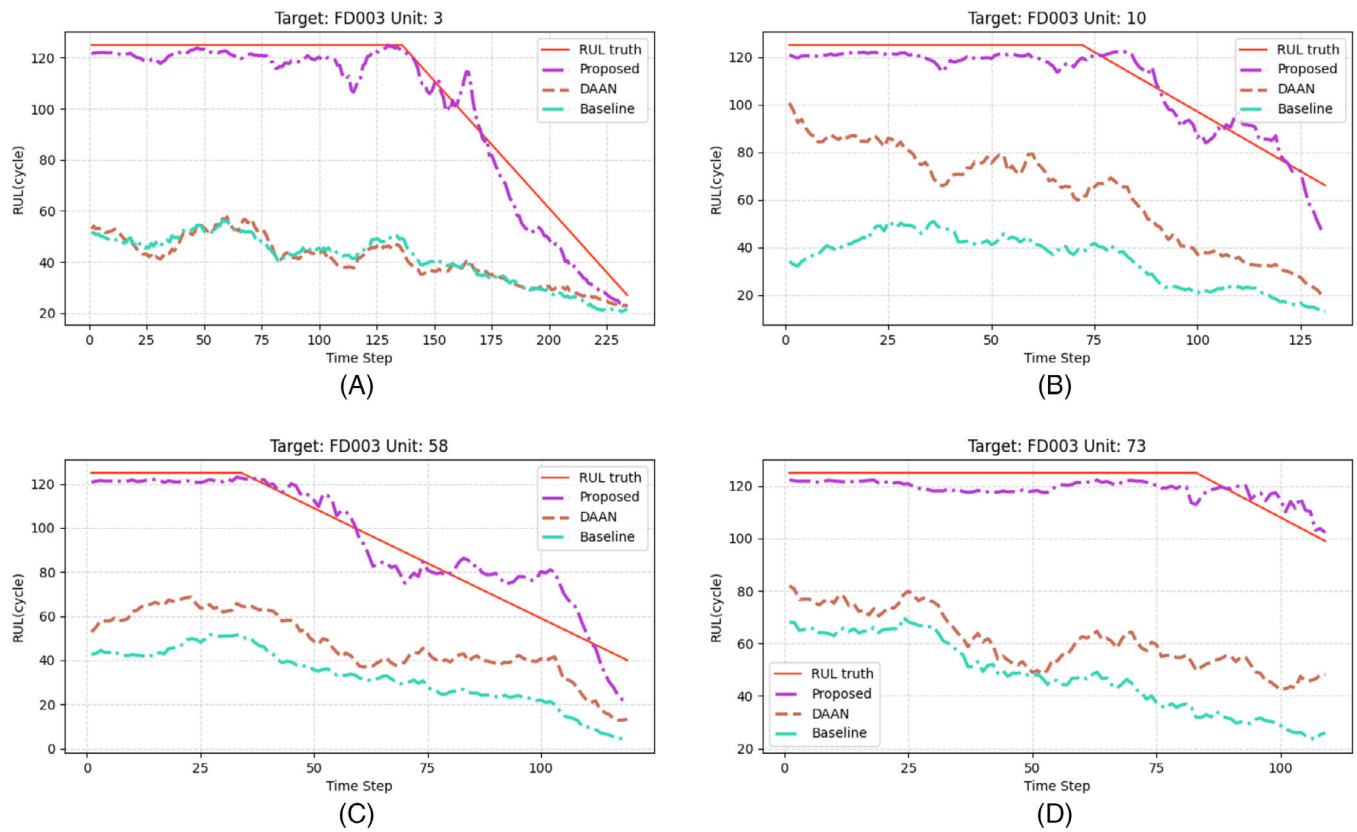


FIGURE 10 Prediction plots of different engine units with different methods on FD1-3.

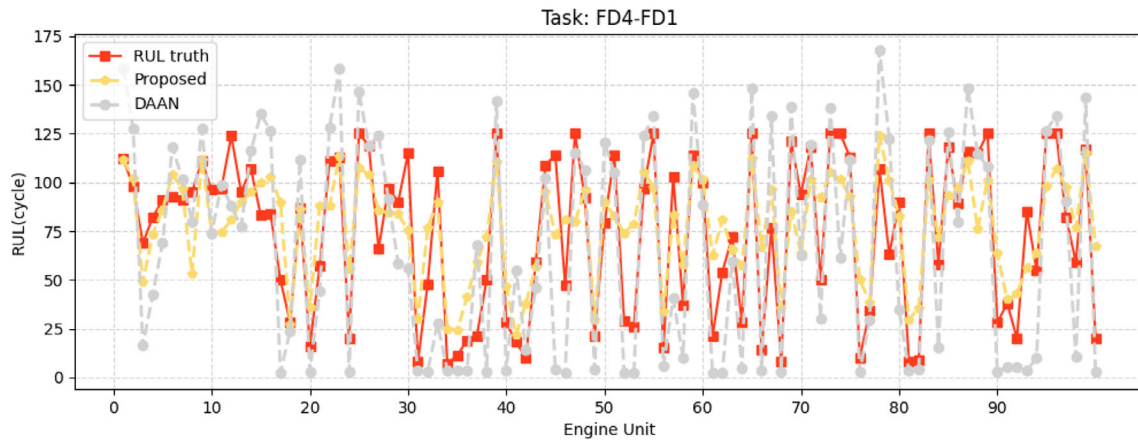


FIGURE 11 RUL prediction results of the USDAN and DAAN methods on the transfer task FD4-1.

the DAAN and USDAN methods on the transfer tasks FD4-1 and FD2-4. The horizontal axis in the figure represents the number of engine units, and the vertical axis represents the RUL of the engine unit. Yellow and gray vertices indicate the RUL predicted values obtained by the USDAN and DANN methods using the multisensor data from the last time period of the test engine unit as input. Red vertices represent the true RUL values of the test engine unit. It is obvious that the yellow line of the USDAN method is closer to the red line representing the true value than the gray line of the DAAN method.

To demonstrate the advantages of our proposed method, this study contrasts the experimental results of other state-of-the-art methods based on the C-MAPSS dataset, as shown in Table 4. Compared with these four methods, the mean value of the RMSE of our proposed method is significantly reduced. Therefore, the overall prediction performance of the proposed USDAN method outperforms these methods. The LSTM-DAAN and CNN-DAAN methods are also adversarial-based domain adaptation approaches, and our proposed method differs from them by the composition of the feature

TABLE 4 RMSE of various transfer tasks for different transfer learning methods on the C-MAPSS dataset.

Transfer Task	Proposed	LSTM-DAAN	CNN-DAAN	TL-DRR	CNN-MMD	DSAN	DAN
FD2-1	24.7	28.1	47.4	34.6	42.4	27.5	23.1
FD3-1	16.2	31.7	33.8	18.3	22.1	14.4	19.1
FD4-1	20.4	31.5	67.4	35.3	37.2	26.5	32.7
FD1-2	39.9	48.6	50.3	43.4	49.4	43.6	48.3
FD3-2	48.7	44.6	68.2	48.8	62.1	44.4	49.9
FD4-2	16.4	24.9	39.1	24.6	25.2	17.1	15.9
FD1-3	19.1	45.9	42.8	24.5	28.3	35.3	40.8
FD2-3	35.5	37.5	40.8	43.2	43.6	41.4	43.1
FD4-3	37.3	27.8	45.5	38.7	46.4	34.7	53.1
FD1-4	43.9	43.8	87.0	45.1	52.5	52.2	47.3
FD2-4	32.1	31.8	38.0	33.1	37.3	36.6	40.0
FD3-4	53.1	47.9	73.7	52.5	64.0	45.4	51.8
Mean	32.3	37.0	52.8	36.8	42.5	34.9	38.8

The bold values represent the best results.

generation network and the application of multilinear conditioning. This also shows that the feature generation network composed of TCN is more conducive to extracting the degradation features of the time series, and the subdomain adversarial based on multilinear conditioning limits the RUL prediction value of different subdomains to the RUL range of the current subdomain to a certain extent, thus avoids the large change of RUL prediction value, and thus reduces the prediction error. TL-DRR and CNN-MMD are domain adaptation approaches based on the multikernel maximum mean differences. Compared with their experimental results, the RMSE of USDAN decreased by 13.9 and 31.6%, respectively. The prediction accuracy of USDAN is higher than DSAN and DAN, which indicates that the unsupervised subdomain adversarial method proposed in this paper can better learn domain-invariant feature representation compared with traditional domain or subdomain adaptation, which improves the accuracy of the mapping relationship. Moreover, the prediction performance of DSAN in these transfer learning methods is second only to USDAN, which also demonstrates that the study of subdomains helps to improve the accuracy and generalization of prediction models. It should be noted that the USDAN method has poor performance on the subdataset FD004. Therefore, in future work, we should focus on the overall performance while also paying attention to the accuracy of RUL prediction under different operating conditions, especially for devices with complex operating settings. For different cross-domain RUL prediction tasks, the parameters of the model can be appropriately adjusted to accommodate different target domains. In addition, the TEL-Bi-LSTM method⁴⁰ studied by Tian et al. uses multiple data sources under multiple working conditions as the source domain, treats the data from each condition as the subsample domain with its own probability distribution, and quantifies the difference between the subsample domain and the online target samples. Moreover, a new transfer ensemble learning framework is proposed, using the domain difference index of submodels and RUL prediction value to achieve the online prediction of target samples with unknown working conditions. The USDAN method mainly explores the influence on the prediction performance of the target condition RUL when the source domain data are only from one operating condition and subdivides multiple subdomains to further learn the domain invariant features of the source and target. The application of multiple source domains in TEL-Bi-LSTM method is worth our further study and research, which can be used as our next research direction.

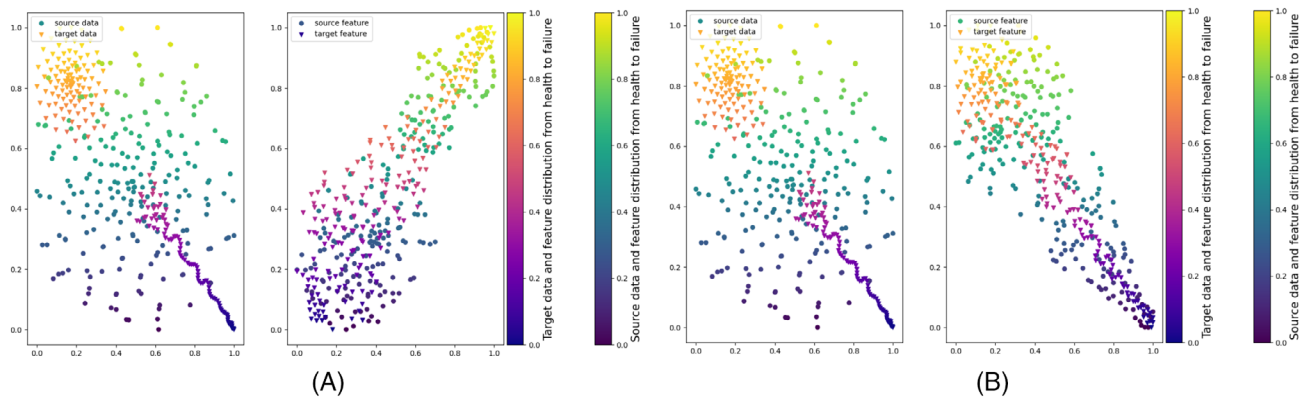
4.4 | Ablation experiment

In order to comprehensively analyze the impact of various key components of USDAN on prediction performance, we designed a set of ablation experiments, including USDAN(with CNN), UDAN(TCN + DA), and DAAN. Among them, USDAN(with CNN) uses the convolutional neural network as the component of the feature extractor, mainly to study whether the TCN method is more favorable for the feature extraction of temporal data than CNN. UDAN(TCN + DA) is a measure-based domain adaptation method, which uses the maximum mean difference to calculate the feature distribution discrepancies between the source and the target, instead of the domain discriminator in USDAN, to promote the extraction of domain invariance. The purpose of designing this method is to explore the prediction performance of domain adaptation

TABLE 5 Performance comparison of the ablation experiments on various transfer tasks.

Ablation experiments	USDAN(with CNN)		UDAN(TCN + DA)		DAAN		USDAN(proposed)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
FD2-1	25.5	20.7	23.1	18.0	33.2	26.7	24.7	19.7
FD3-1	17.6	14.6	19.1	14.1	17.8	14.0	16.2	13.1
FD4-1	28.7	23.9	32.7	25.2	30.4	26.4	20.4	17.4
FD1-2	41.4	36.3	48.3	41.0	43.4	37.6	39.9	35.0
FD3-2	45.8	39.6	50.0	40.8	51.4	41.9	48.7	41.3
FD4-2	18.1	14.2	15.9	11.7	17.4	13.0	16.4	12.7
FD1-3	38.8	34.4	40.8	34.7	45.5	38.7	19.1	14.8
FD2-3	48.1	39.1	43.1	34.7	39.6	32.8	35.5	31.0
FD4-3	32.3	24.1	53.1	44.3	51.7	42.7	37.3	28.6
FD1-4	42.7	38.1	47.3	41.9	45.3	39.0	43.9	38.9
FD2-4	28.2	23.1	40.0	31.6	37.3	28.9	32.1	26.5
FD3-4	45.4	39.6	51.8	43.0	53.5	45.0	53.1	44.3
Mean	34.4	29.0	38.8	31.8	38.9	32.2	32.3	26.9

The bold values represent the best results.

**FIGURE 12** Visualization of domain-invariant features. (A) USDAN, (B) DAAN.

methods based on adversarial and metrics. DAAN has been comparative analyzed in the previous section and will not be elaborated here. As can be seen from Table 5, the prediction performance of USDAN is better than these three methods, followed by USDAN (with CNN). The experimental results show that the study of the subdomain problems is crucial. Moreover, both the design of the model framework and the proposal of subdomain confrontation in this paper improve the performance of RUL prediction.

4.5 | Feature visualization and complexity analysis

To further demonstrate the advantages of the proposed method, we used t-SNE visualization to represent the domain-invariant features generated by the USDAN method by using the principal component analysis method with dimension reduction and compared it with the DAAN method. As shown in Figure 12, the gradient color indicates the degradation states, and as the color gradually deepens, the degradation gradually intensifies. The left side of the graph represents the distribution of raw data in the source and target domains, while the right side represents the distribution of degradation features of the source and target domains. It is obvious from the figure that the distribution of triangles and circles with the same color depth is more similar in Figure 12A than in Figure 12B, which indicates that the domain invariant features of the source domain and target domain generated by USDA are more closely aligned within the same subspace. However, the visual distribution of domain-invariant features extracted by DAAN shown in Figure 12B is globally roughly aligned, but quite different in the subdomain.

TABLE 6 The comparison of the time complexity.

Methods	USDAN	DAN	DSAN
Time cost(s)	25	34	40

Furthermore, we also analyzed the complexity of the method from the training time perspective for USDAN, DAN, and DSAN. During the model training process, we recorded the average consumption time of each iteration of the three methods. As shown in Table 6, the time complexity of these three methods is ordered from low to high: USDAN, DAN, and DSAN. Because both DAN and DSAN require more computational effort for the feature distribution differences of the source and target domain. Moreover, DSAN needs a difference measure for each subdomain separately, so its time complexity is the highest among the three methods. Overall, although DSAN and USDAN can achieve similar prediction effects, USDAN performs better in computational complexity.

5 | CONCLUSIONS

In this paper, we propose an USDAN using a labeled dataset as the source domain and an unlabeled dataset as the target domain to complete RUL prediction tasks under various different operating conditions. In contrast to the traditional DAAN, our proposed USDAN adds a subdomain clustering network to partition the degeneration features of the whole lifetime into multiple subdomains. Moreover, a multilinear conditioning method is designed to combine the degeneration features with the corresponding subdomain category labels. With the limitation of subdomain labels, the domain-invariant features generated by the feature generation network have a higher correlation with the labels during the model optimization process, which helps to reduce the error between the RUL labels obtained by the label prediction network and the true value. Therefore, USDAN solves the problem of decreased prediction accuracy due to domain mismatch and achieves better performance than other transfer learning methods. In addition, USDAN uses temporal convolutional blocks to construct the feature generation network, enabling the extracted features to more comprehensively represent the degeneration states of multisensor data. Because the TCN combines the advantages of the traditional CNN and RNN, the extracted features are both temporal and spatial.

However, the method proposed here still has some limitations, which assume that the trained model uses degradation data with run-to-failure full life cycle data. This is also a prerequisite for most transfer learning methods. However, in the actual application scenario, the collected data are likely to be incomplete due to the operating environment or sensor failure. Therefore, our future studies will focus on completing the cross-condition RUL prediction task using incomplete degradation data.

AUTHOR CONTRIBUTIONS

Methodology: Zhenfei Wen and Yi Lyu. *Validation:* Zhenfei Wen. *Writing—original draft:* Zhenfei Wen. *Conceptualization:* Yi Lyu. *Writing—review and editing:* Yi Lyu. *Formal analysis:* Aiguo Chen.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in Turbofan Engine Degradation Simulation at <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoc/pcoc-data-set-repository/>. These data were derived from the following resources available in the public domain: - C-MAPSS, <https://data.nasa.gov/Aeospace/CMAPSS-Jet-Engine-Simulated-Data/ff5v-kuh6>.

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